

The empirical recurrence for OEIS A303186, recovered from its tail

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Abstract

OEIS A303186 records “Empirical recurrence of order 84” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 250$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 7$, so the recurrence is proved for every $n \geq 91$. Its last coefficient is $c_{84} = -8 \neq 0$, so no further tail satisfies anything shorter; any order-84 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A303186 is “Number of $6 \times n$ 0..1 arrays with every element equal to 0, 1, 2, 5 or 6 horizontally, diagonally or antidiagonally adjacent elements, with upper left element zero.”. It has offset 1 and begins

32, 857, 1938, 5050, 29124, 218980, 1064399, 5463686, ...

The entry states:

Empirical recurrence of order 84 (see link above)

As of the “Last modified” line on the live entry (Apr 19 2018 13:28:23, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 250$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 250$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 84: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 7$ is the first whose minimal order is exactly the stated 84.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 500$ exact terms from $n = 7$ on, it returns order 84 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{84} c_i a(n-i),$$

c_1	8	c_{22}	−16552	c_{43}	1317854	c_{64}	−34719
c_2	−15	c_{23}	−218378	c_{44}	3599621	c_{65}	90066
c_3	22	c_{24}	85355	c_{45}	−1958442	c_{66}	−140737
c_4	−84	c_{25}	23221	c_{46}	−1146746	c_{67}	3664
c_5	−128	c_{26}	−399015	c_{47}	2788840	c_{68}	19471
c_6	662	c_{27}	112408	c_{48}	−883853	c_{69}	44907
c_7	396	c_{28}	744286	c_{49}	190618	c_{70}	−41139
c_8	−2404	c_{29}	−1680413	c_{50}	1483962	c_{71}	30905
c_9	1146	c_{30}	498680	c_{51}	−1726812	c_{72}	−817
c_{10}	325	c_{31}	2630295	c_{52}	−110998	c_{73}	−3285
c_{11}	2603	c_{32}	−1692176	c_{53}	206998	c_{74}	10186
c_{12}	−535	c_{33}	437231	c_{54}	−230310	c_{75}	8249
c_{13}	649	c_{34}	2385164	c_{55}	−571651	c_{76}	2881
c_{14}	−10836	c_{35}	−1896296	c_{56}	−360045	c_{77}	3592
c_{15}	−12089	c_{36}	−615356	c_{57}	118331	c_{78}	2932
c_{16}	86016	c_{37}	730307	c_{58}	−185935	c_{79}	331
c_{17}	−118009	c_{38}	−2116378	c_{59}	103289	c_{80}	26
c_{18}	19884	c_{39}	−3936161	c_{60}	−263945	c_{81}	−2
c_{19}	103147	c_{40}	−354788	c_{61}	140541	c_{82}	−146
c_{20}	−235451	c_{41}	1605761	c_{62}	17432	c_{83}	−24
c_{21}	315357	c_{42}	1094768	c_{63}	−97708	c_{84}	−8

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 91$, and it is the recurrence OEIS A303186 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 7$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{84} - c_1x^{84-1} - \dots - c_{84}$. Its constant term is $-c_{84} = 8$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 84 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 84, so $\deg q = 84$, and it is monic. It holds from some index t on. From $\max(t, 7)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 84, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 19 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 84, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -8 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-84 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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